

Celebal Assignment – Week 2

Overview

This repository contains solutions for Week 2 completed as part of my internship practice work. The assignment focuses on fundamental and intermediate SQL concepts including data retrieval, filtering, constraints, indexing, aggregation, grouping, and query optimization.

Topics Covered

- SELECT Queries
- WHERE Clause Filtering
- DISTINCT Keyword
- Primary Keys & Constraints
- UNIQUE, NOT NULL, CHECK Constraints
- Aggregate Functions (COUNT, SUM, AVG, MAX, MIN)
- GROUP BY & HAVING Clauses
- ORDER BY Sorting
- Date Filtering
- Indexing & Query Optimization
- SARGable Queries

Key Learning Outcomes

Through this assignment, I gained practical experience in:

- Writing efficient SQL queries
- Understanding relational database design
- Applying constraints to maintain data integrity
- Using indexes to improve query performance
- Performing data analysis using aggregate functions
- Optimizing queries for better scalability

Technologies Used

- SQL
- MySQL / Relational Database Concepts